

Name: Aiman Imran

Seat # 23122096

Semester: VI

Section: B

Subject: AI

Teacher: Ma'am Uzma

CineMind

(AI Powered Movie Recommendation Agent)

Agent Name: Movie Recommendation Agent

Problem Formulation in Context of an Agent:

Environment: CineMind operates in a movie database containing movies, genres, ratings, languages, and user preferences.

Sensors: The agent receives the user's search, preferred genre, language, mood, ratings, and watch history.

Knowledge Base: The knowledge base stores movie titles, genres, actors, ratings, languages, descriptions, and user preferences.

Algorithm/Reasoning: The agent analyzes the user's preferences and compares them with movie information to find and rank the most suitable movies.

Actuator / Rational Action: The agent displays personalized movie recommendations along with their details and reasons for recommending them.

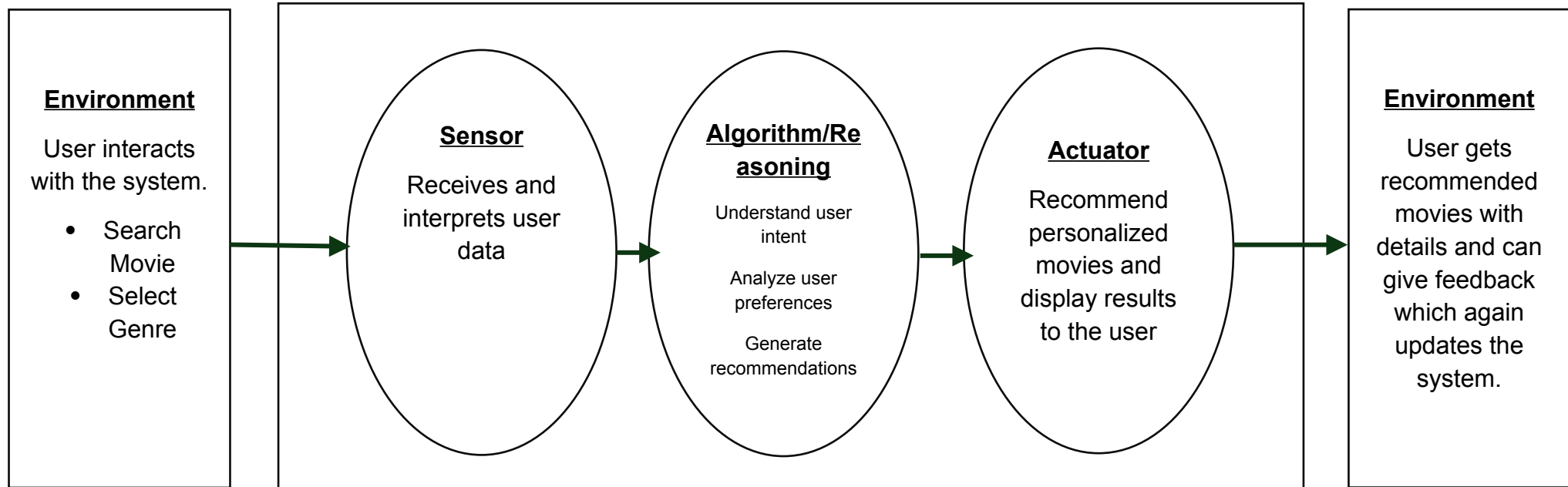
User	Genre	Mood	Language	Movie	Condition	Result
18+	Sci-Fi	Curious	English	Interstellar	If genre = Sci-Fi AND mood = Curious	Recommended
18+	Action	Excited	English	The Dark Knight	If genre = Action AND mood = Excited	Recommended
13+	Comedy	Happy	Hindi	Three Idiots	If genre = Comedy AND mood = Happy	Recommended
All	Animation	Family	English	Toy Story	If genre = Animation AND mood = Family	Recommended
18+	Thriller	Suspense	English	Inception	If genre = Thriller AND mood =	Recommended

					Suspense	
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Problem Formulation:

Recommend a suitable movie to the viewer based on their age group, favorite genre, preferred language, mood, and watched history.

Graphical Representation:



Evaluation in the Context of PEAS:

Performance Measure: Recommendation relevance/accuracy, user satisfaction, click-through and watch-through rate, diversity of suggestions, speed of response.

Environment: The user's profile and preference history, the movie catalog (genre, rating, cast, duration, release year), and the app/streaming interface through which the user interacts and receives suggestions.

Sensor: Display/output module that renders the ranked movie list, thumbnails, trailers, and the match-reason text shown to the user.

Actuator: Input/intake module that senses (reads) the user's search queries, star ratings, watch history, selected mood/genre filters, and browsing/click behavior.

Environment Properties in Context of Agent:

1. Observability – Partially Observable

CineMind can only see the information that the user provides, such as their favorite genres, ratings, and watch history. It cannot know everything about the user's taste or mood, so the environment is **partially observable**.

2. Determinism – Deterministic

CineMind gives a predictable result when the same user information and movie data are provided. Since the basic recommendation rules do not involve randomness, the environment is **deterministic**.

3. Episodic vs. Sequential – Episodic

Each movie recommendation request is treated as a separate task. One recommendation does not necessarily depend on the previous recommendation, so the environment is **episodic**.

4. Static vs. Dynamic – Semi-Static

The movie database and user information normally remain unchanged while CineMind is making a recommendation. However, new movies, ratings, or user preferences can be added later, so the environment is **semi-static**.

5. Discrete vs. Continuous – Discrete

CineMind works with separate values such as movie genres, languages, ratings, and age groups. These values are clearly defined rather than continuously changing, so the environment is **discrete**.

6. Known vs. Unknown – Known

The movie information and recommendation rules are already stored in the system. Therefore, CineMind knows the rules it uses to make recommendations, making the environment **known**.

7. Single-Agent vs. Multi-Agent – Single-Agent

CineMind works as a single intelligent agent that provides recommendations to the user. It does not need to compete or cooperate with other intelligent agents, so it is a **single-agent environment**.